

Multiplierless Reconfigurable Processing Element for Mixed Radix-2/3/4/5 FFTs

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Abstract—This paper presents area-efficient building blocks for computing fast Fourier transform (FFT): multiplierless processing elements to be used for computing of radix-3 and radix-5 butterflies and reconfigurable processing element supporting mixed radix-2/3/4/5 FFT algorithms. The proposed processing elements are based on Wingorad Fourier transform algorithm. However, multiplication is performed by constant multiplier instead of a general complex-valued multiplier. The proposed process elements have potential use in both pipelined and memory based FFT architectures, where the non-power-of-two sizes are required. The results show that the proposed multiplierless processing elements reduce the significant hardware cost in terms of adders.

Index Terms—Fast Fourier transform (FFT), Memory-based FFT, Mixed radix

I. INTRODUCTION

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In today's digital signal processing (DSP) systems, there is often need to transform a signal between time and frequency domains. For this reason, the discrete Fourier transform (DFT) has become one of the most important algorithms; however, this technique is computationally intensive. A number of algorithms have been proposed for efficient and fast computation of DFT, collectively recognized as fast Fourier transforms (FFT) [1]–[4].

FFT gained significant attention in communications systems because it is an important operation in orthogonal frequency division multiplexing (OFDM) systems. OFDM is a leading modulation technique which adequately broaden channel usage and abridge inter-symbol interference (ISI) along with inter-carrier interference (ICI) generated by multi path effects. It is mainly used in digital audio broadcasting (DAB), digital video broadcasting-terrestrial (DVB-T) and digital video broadcasting-handheld (DVB-H). In certain OFDM based communication applications such as 3GPP LTE, FFT is employed to spread the transmitted data over a number of subcarriers on different frequencies.

Traditionally FFTs of size of a power-of-two have been exploited as design of non-power-of-two FFT processors is more challenging as both data management and data processing is irregular [5]–[9]. Therefore, most of the research work on FFT algorithms and architectures has focused on power-of-two DFT sizes. However, there are certain applications, where non power-of-two FFT sizes are required. There are certain algorithms for channel estimation, which utilize the available

subcarriers, demand FFT sizes of 1200 or 2400 points. Furthermore, variable-length FFT architectures are required, for instance, 3GPP LTE demands various FFT sizes such as 128, 256, 512, 1024, 1536, 2048 and 12 – 1296, respectively [7], [10]–[14].

In general, literature classify the FFT processors into two architectures: memory-based architectures and pipelined architectures [15]–[18]. On the one hand, memory-based FFT architectures provides low power and small area solution; however, it suffers from a long latency. There are two important challenges during its design: one is the conflict-free memory access and other is the design of core for computation of butterflies (small point DFTs or radices). The design of the core for a single radix is relatively simple and well known topic. However, the design of mixed-radix architectures still proposes some challenges. Usually, mixed-radix architectures are used for hardware cost reduction purposes, which is achieved by reusing the adders and multipliers [5], [7], [10]. On the other hand, pipelined architectures are popular in real-time applications, which requires high throughput. However, these architectures have larger hardware cost. In a similar fashion, the design of non-power-of-two pipelined architectures is a challenging task.

In this paper, we propose multiplierless processing elements for computing butterfly operations in non-power-of-two FFTs. These processing elements can be used in both pipelined and memory-based architectures. The proposed processing elements do not require expensive general complex-valued multipliers as they are replaced with constant multipliers realized with cheap shift-and-add circuits. The independent architectures of radix-3 and radix-5 can be used in pipelined architectures; however, reconfigurable processing element can be used in memory based FFT architectures.

The rest of paper is organized as follows. We briefly review the background on the individual architectures of radix-2/3/4/5 in section II. Then, we explain the main idea behind the multiplierless multiplication in fixed-point arithmetic in section II-C. Then, section III presents the proposed multiplierless architectures. In section IV we compare the hardware cost of architectures. Finally section V provides the main conclusions of the paper.

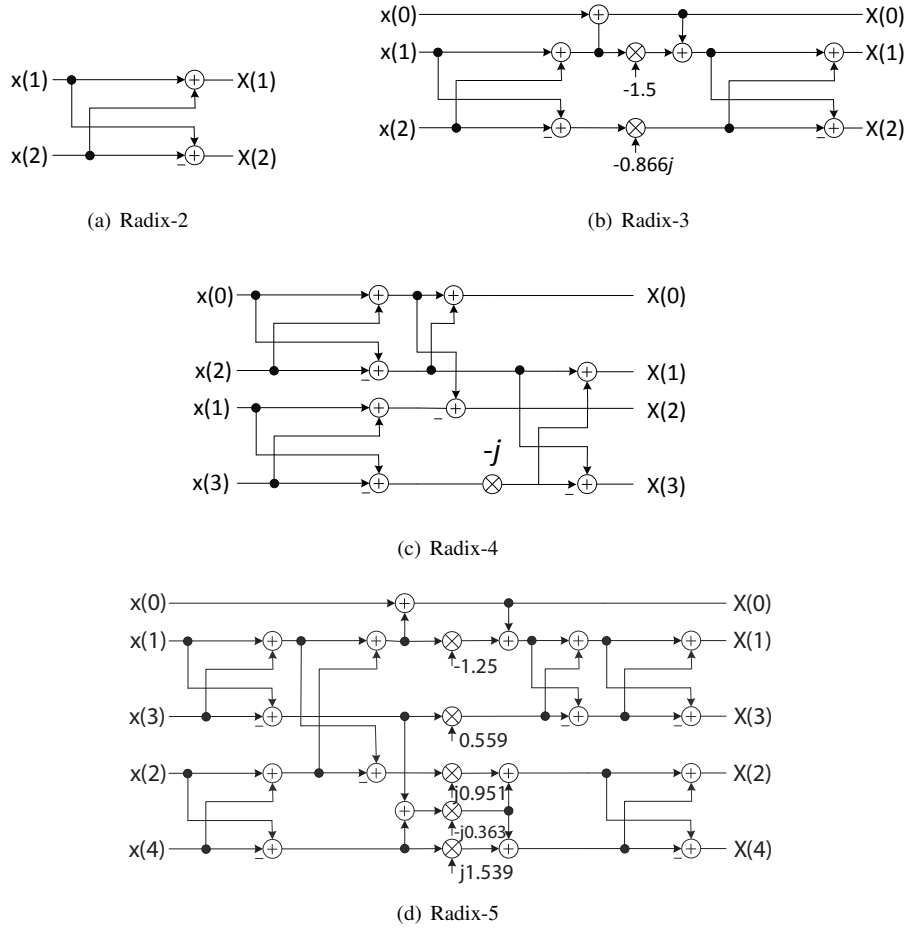


Fig. 1. Signal flow graph of radix-2/3/4/5.

II. RELATED WORK

A. Radix-2/3/4/5 FFT

FFT algorithms are typically composed of smaller DFTs. If the FFT consist of DFT's of size r , the FFT is called as radix- r FFT. The radix is one of most important modules in implementation of FFT algorithms, also called as a butterfly. It is implemented directly, where the number of arithmetic components are equal to the number of arithmetic operations. Likely, this is isomorphic mapping of the radix flow graph on the hardware.

The radix-2 FFT can be expressed as;

$$\begin{bmatrix} X(0) \\ X(1) \end{bmatrix} = F_2 \cdot \begin{bmatrix} x(0) \\ x(1) \end{bmatrix}; F_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad (1)$$

where F_2 is the transform matrix for 2-point DFT and $x(\cdot)$ and $X(\cdot)$ represent the input and output sequences, respectively.

In a similar fashion, the basic building blocks for radix-3, radix-4, and radix-5 FFTs are the corresponding DFTs, i.e.:

$$F_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -0.5 - 0.866i & -0.5 + 0.866i \\ 1 & -0.5 + 0.866i & -0.5 - 0.866i \end{bmatrix} \quad (2)$$

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix} \quad (3)$$

$$F_5 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0.30 - 0.95i & -0.81 - 0.58i & -0.81 - 0.58i & 0.31 + 0.95i \\ 1 & -0.81 - 0.58i & 0.31 + 0.95i & 0.31 - 0.95i & -0.81 + 0.58i \\ 1 & -0.81 + 0.58i & 0.31 - 0.95i & 0.31 + 0.95i & -0.81 - 0.58i \\ 1 & 0.31 + 0.95i & -0.81 + 0.58i & -0.81 - 0.58i & 0.31 - 0.95i \end{bmatrix}. \quad (4)$$

The signal flow graphs (SFG) that represent the basic computations of radix-2, radix-3, radix-4, and radix-5 FFTs are shown in Fig. 1. The SFG of radix-2 and radix-4 are based on the direct implementation of DFT algorithm, whereas the signal flow graphs of the radix-3 and radix-5 are based on Winograd Fourier transform algorithm (WFTA) discussed in the following section.

B. Winograd Fourier Transform Algorithm

This algorithm has the minimum number of multiplications at the expense of introducing a few extra additions [2]. Although WFTA is very efficient for small prime size DFTs, for larger sizes the number of additions becomes quickly unpractical. In the WFTA, the DFT equation is written as

$$[X(0) \dots X(N-1)]^T = O \cdot M \cdot I \cdot [x(0) \dots x(N-1)]^T, \quad (5)$$

TABLE I
DIFFERENT NON-TRIVIAL COEFFICIENTS AND TOTAL NUMBER OF ADDERS TO REALIZE THE MULTIPLIERLESS STRUCTURE FOR RADIX-5.

Fractional Bits	Coefficients	Number of Adders
8	$\frac{320}{256}, \frac{143}{256}, \frac{243}{256}, \frac{394}{256}, \frac{93}{256}$	11
9	$\frac{640}{512}, \frac{286}{512}, \frac{487}{512}, \frac{788}{512}, \frac{186}{512}$	11
10	$\frac{1280}{1024}, \frac{572}{1024}, \frac{974}{1024}, \frac{1576}{1024}, \frac{372}{1024}$	11
11	$\frac{2560}{2048}, \frac{1145}{2048}, \frac{1948}{2048}, \frac{3152}{2048}, \frac{774}{2048}$	12
12	$\frac{5120}{4096}, \frac{2290}{4096}, \frac{3896}{4096}, \frac{6303}{4096}, \frac{1488}{4096}$	12
13	$\frac{10240}{8192}, \frac{4579}{8192}, \frac{7791}{8192}, \frac{12606}{8192}, \frac{2976}{8192}$	14
14	$\frac{20480}{16384}, \frac{9159}{16384}, \frac{15582}{16384}, \frac{25212}{16384}, \frac{5952}{16384}$	14
15	$\frac{40960}{32768}, \frac{18318}{32768}, \frac{31164}{32768}, \frac{50425}{32768}, \frac{11904}{32768}$	15

where I is a matrix that corresponds to additions between inputs, M is a diagonal matrix with the multiplications and O is a matrix corresponding to additions after the multiplications. These algorithms require multiplication by real or imaginary values.

Denote $c = r + j \cdot i$ to be a complex constant only with real part $r = \cos \alpha, i = 0$ and $x = x_{re} + j \cdot x_{im}$ be a complex variable. Then, the complex output $Y = Y_{re} + Y_{im}$ is computed with

$$\begin{bmatrix} Y_{re} \\ Y_{im} \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 \\ 0 & \cos \alpha \end{bmatrix} \begin{bmatrix} x_{re} \\ x_{re} \end{bmatrix}, \quad (6)$$

where imaginary part is zero. Conversely, multiplication of x by a constant c with only imaginary part $r = 0, i = \sin \alpha$ is described as

$$\begin{bmatrix} Y_{re} \\ Y_{im} \end{bmatrix} = \begin{bmatrix} 0 & -\sin \alpha \\ \sin \alpha & 0 \end{bmatrix} \begin{bmatrix} x_{re} \\ x_{re} \end{bmatrix}. \quad (7)$$

These considerations are based on the WFTA and permit to transform (2) into (5). Therefore, WFTA requires only a semi-complex multiplier for multiplication with complex data and provides two times smaller hardware cost.

C. Fixed-Point Arithmetic

Ideally, a multiplication of a complex-valued number with a constant should be represented with infinite precision. However, in modern digital systems, the memory registers with finite capacity are used. Therefore, finite-precision arithmetic is used to represent values and perform computations.

Consider 2's complement fixed-point arithmetic [19], real numbers are represented as integers scaled by an implicit quantization factor. Such arithmetic benefits from fast bit operations. However, some accuracy concerns are related to such passage to finite precision.

Let r and i be the real and imaginary parts of the complex constant c . In our case $r < 1$ and $i < 1$, i.e., we have only the fractional part to be represented. Therefore, if implemented with b bits, the real and imaginary parts are stored as integers $R, I \in \mathbb{Z} \in [-2^{-b+1}, -2^{-b+1} - 1] \cap \mathbb{Z}$, which are obtained by

$$R = \lfloor 2^{-b+1} \cdot \cos \alpha \rfloor, \quad I = \lfloor 2^{-b+1} \cdot \sin \alpha \rfloor, \quad (8)$$

where $\lfloor \cdot \rfloor$ represents the rounding operation (in our case, it is truncation). Therefore, the fixed-point counterparts of r and

i are $R \cdot 2^{-b+1}$ and $I \cdot 2^{-b+1}$, respectively. The quantization error ε depends on the chosen rounding mode. In case of the truncation, we obtain

$$-2^{-b+1} \leq \varepsilon < 0. \quad (9)$$

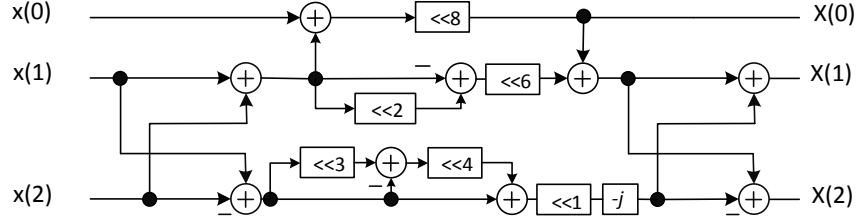
III. PROPOSED ARCHITECTURE

The architecture for radix-2/3/4/5 computations is obtained by direct implementation of the SFG from Fig. 1, which requires a number of hardware components equal to arithmetic operations.

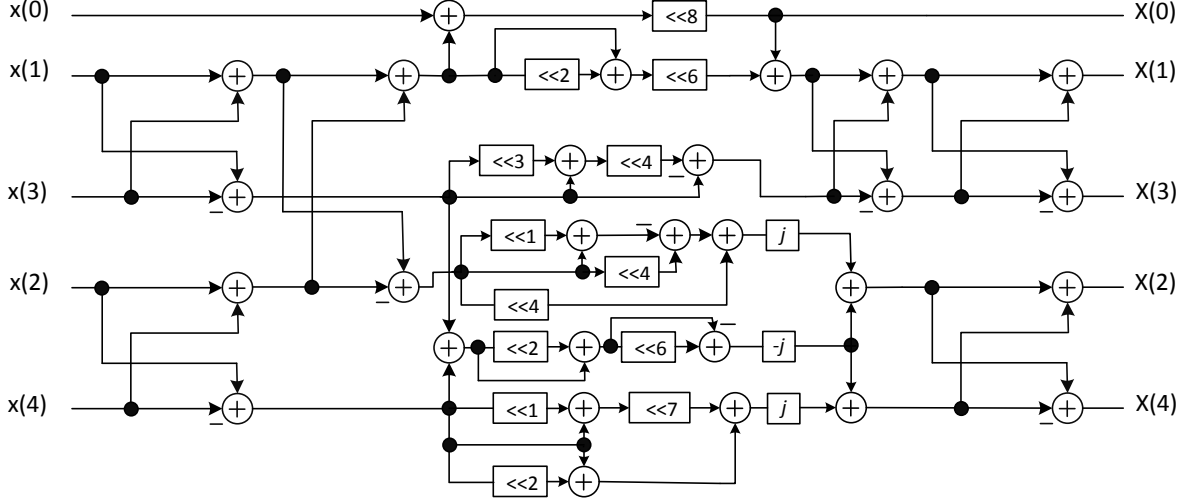
A. Multiplierless Radix-3/5 WFTA Architecture

A number of multiplications by a constant is used in radix-3/5 WFTA architecture. It is possible to use shift-and-add circuit to efficiently implement the multiplication instead of a general complex-valued multiplier. The design of shift-and-add circuit depends on the selection of coefficients and shift-and-add implementation. Traditionally, the coefficients are obtained by the rounding of pre-computed constants; however, the canonical signed digit representation is used to reduce the number of non-zero digits with respect to the simple binary representation, which reduces the number of adders [20], [21]. Further simplification is achieved by single constant multiplication techniques [22], [23]. They exploit the redundancy in the multiplication by a single constant. Additionally, improvement in accuracy and complexity can be achieved by addition-aware coefficient quantization method [24].

Using these techniques proposed in [20]–[23], [25], we obtain a number of candidate coefficients whose fractional bits range are tabulated in Table I. The selection is based on specifications like adder cost, which has relationship with the coefficient quantization error ε . In this work, we selected 8-bit coefficients, which have required the minimum number of adders; however, the length can be increased according to accuracy requirements of the design at hand. The circuits will be changed according to the selection of the word length. The



(a) Radix-3



(b) Radix-5

Fig. 2. Multiplierless architectures for radix-3 and radix-5 FFTs.

8-bit coefficients circuit can be expressed as:

$$\begin{aligned}
 320 \cdot X &= ((A \cdot 4 + A) \cdot 64); \\
 143 \cdot X &= (((X \cdot 8 + X) \cdot 16) - X); \\
 243 \cdot X &= (X \cdot 256 - (X \cdot 16 - (X \cdot 2 + X))); \\
 393 \cdot X &= ((X \cdot 2 + X) \cdot 128 + (X \cdot 8 + X)); \\
 93 \cdot X &= ((X \cdot 4 - X) \cdot 8 - (X \cdot 4 - X)). \quad (10)
 \end{aligned}$$

Usually, the shifts are free of cost as they are hardwired. The final multiplierless architectures of radix-3/5 are illustrated in Fig. 2.

B. Multiplierless Reconfigurable Radix-2/3/4/5 FFT

The multiplierless reconfigurable processing element is based on mapping the 2/3/4/5-radix onto a single processing core. We reuse the hardware circuit by multiplexers for computation of 2/3/4/5-radix. The complexity of circuit is equal to radix-5 with a few additional multiplexers. The multiplierless radix-5 WFTA requires the most adders. Further, main challenge is to reduce the number of multiplexers. To obtain this, it is important to find common parts in the signal flow graphs that can be mapped without multiplexers. In addition,

multiplexer can be avoided by setting the unused inputs of the circuit to zero, which removes the unnecessary connections of the circuit. Using these techniques, a solution with only nine two-to-one multiplexers controlled with two control signals, has been designed. The resulting architecture is shown in Fig. 3. The input and output relations are tabulated in Table II, where the dashes denote "don't care" conditions and 0 denotes that the input should be zeroed for proper operations. Finally, signals controlling the multiplexers are shown in Table III, where dash denotes "don't care" condition.

IV. RESULTS

In the proposed processing elements, a general complex-valued multiplier is replaced by a constant multiplier, i.e., a circuit that carries out a multiplication by a single constant [26]. The constant multiplier circuit is designed based on shift-and-add circuit. As a result, the proposed individual and reconfigurable architectures consist of adders/subtractor. However, additional multiplexers are required in reconfigurable circuit.

Considering the equivalent cost of semi-complex multiplier implemented by using a general complex-valued multiplier: Booth encoding and multiplexer, which are equal to 16 and

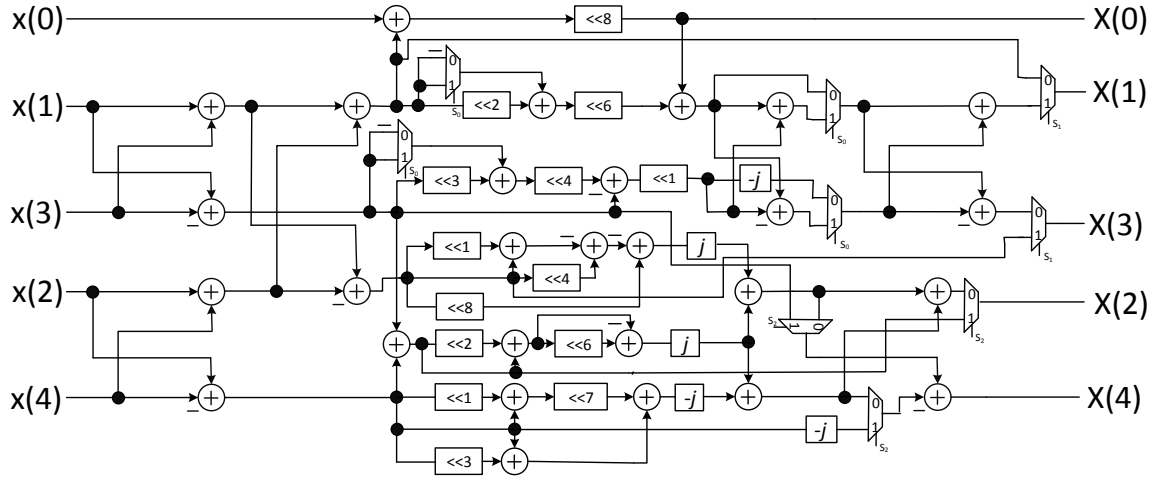


Fig. 3. Multiplierless reconfigurable architecture for radix-2/3/4/5 FFT.

TABLE II
INPUTS AND OUTPUTS OF PROPOSED ARCHITECTURE.

Input configurations					Output configurations				
Index IN	DFT size				Index OUT	DFT size			
	2	3	4	5		2	3	4	5
0	0	x(0)	0	x(0)	0	-	X(0)	-	X(0)
1	x(0)	x(1)	x(0)	x(1)	1	X(0)	X(1)	X(0)	X(1)
2	x(1)	x(2)	x(1)	x(2)	2	X(1)	X(2)	X(1)	X(2)
3	0	0	x(2)	x(3)	3	-	-	X(2)	X(3)
4	0	0	x(3)	x(4)	4	-	-	X(3)	X(4)

1/4 adders, respectively [27]. Table IV compares the complexity of the proposed architectures to other architectures. We can observe that the complexity in terms of adder cost of proposed radix-3 and radix-5 multiplierless processing elements is reduced 60% and 70%, respectively. Whereas, for reconfigurable processing element the proposed approach achieves 76% reduction in adder cost.

V. CONCLUSIONS

This paper presents the multiplierless processing elements for designing the non-power-of-two FFTs. The proposed processing elements are based on Winograd Fourier transform algorithm. However, multiplication is performed by a constant multiplier instead of a general complex-valued multiplier. These are used to calculate radix-2, radix-3, radix-4, and radix-5. It is shown that the complexity of radix-3/5 and

reconfigurable mixed-radix 2/3/4/5 is reduced up to 60%, 70%, and 75%, respectively.

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TABLE III
CONTROL SIGNALS TO OBTAIN DIFFERENT DFT SIZES.

DFT size	s_0	s_1	s_2
Radix-2	-	0	0
Radix-3	0	1	-
Radix-4	-	0	1
Radix-5	1	1	0

TABLE IV
HARDWARE COMPLEXITY (EQUIVALENT NUMBER OF ADDERS).

DFT size	Proposed	[5], [7]	Reduction(%)
Radix-3	9	23	60
Radix-5	28	97	70
Mixed radix-2/3/4/5	30.25	130	76

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